UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	CSC429
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions) PART B (1 Question)
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages.

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PART A (80 MARKS)

Answer ALL questions in this part.

QUESTION A1 **(5 marks)**

Computer generations are distinguished by their main technology.

(a) State the main technology of the first and third generations. (2 marks)

(b) Explain ONE major limitation of first-generation computers. (3 marks)

QUESTION A2 **(5 marks)**

The CPU contains a control unit and an arithmetic logic unit.

(a) Explain the function of the control unit. (2 marks)

(b) Explain the function of the ALU. (3 marks)

QUESTION A3 **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) Represent -16 in 8-bit sign-magnitude form. (2 marks)

(b) Represent -16 in 8-bit two's complement form. (3 marks)

QUESTION A4 **(5 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) Convert 0.5 to IEEE 754 single-precision format. Give the result in hexadecimal and show the sign, exponent and mantissa fields. (5 marks)

QUESTION A5 **(10 marks)**

Answer the following data-representation questions. Show your complete workflow.

(a) Calculate 76 - (-27) using 8-bit sign-magnitude. Show the complete workflow and the result in byte form. (5 marks)

(b) Calculate 76 - (-27) using 8-bit two's complement. Show the complete workflow and the result in byte form. (5 marks)

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QUESTION A6 **(10 marks)**

The machine cycle is the basic sequence of CPU operations.

(a) List the phases of the machine cycle. (2 marks)

(b) Explain the role of the PC, IR, MAR, MBR and CU during the fetch phase. (3 marks)

(c) Explain what happens during the execute phase and how the ALU and registers are used. (5 marks)

QUESTION A7 **(10 marks)**

Cache memory stores recently accessed data and instructions.

(a) Define cache hit and cache miss. (2 marks)

(b) Explain spatial locality and temporal locality. (4 marks)

(c) Compare direct-mapped, set-associative and fully associative mapping. (4 marks)

QUESTION A8 **(10 marks)**

Multiple interrupts can arrive while the CPU is servicing another interrupt.

(a) Explain nested interrupts. (3 marks)

(b) Describe how interrupt priority determines the order of service. (3 marks)

(c) Explain the sequence of events when a higher-priority interrupt arrives during an ISR. (4 marks)

QUESTION A9 **(10 marks)**

Interrupts can be maskable or non-maskable.

(a) Differentiate maskable and non-maskable interrupts. (3 marks)

(b) Explain what an interrupt handler (ISR) does. (3 marks)

(c) Describe TWO methods for handling multiple interrupts. (4 marks)

QUESTION A10 **(10 marks)**

Multicore processors are common in modern computers.

(a) Define multicore processing. (2 marks)

(b) Explain TWO benefits of multicore processing. (4 marks)

(c) Describe NUMA and ONE challenge in a distributed-memory system. (4 marks)

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PART B (20 MARKS)

Answer the question in this part.

QUESTION B1 **(20 marks)**

Write appropriate programs using zero-, one- and three-address instructions to compute Z = (A + B) / (C - D). Registers A, B, C, D and E must not be modified; X, Y and Z are temporary registers; the final answer must be stored in Z. Available instructions: zero-address PUSH, POP, ADD, SUB, MUL, DIV; one-address LOAD, STORE, ADD, SUB, MUL, DIV; three-address ADD, SUB, MUL, DIV.

(a) Write a zero-address program for the expression. (6 marks)

(b) Write a one-address program for the expression. (6 marks)

(c) Write a three-address program for the expression. (8 marks)